



# Economics

**Government Intervention  
(tax, Subsidy Maximum Price, Min  
Price)**

**KRISHAN RATHNAYAKE**

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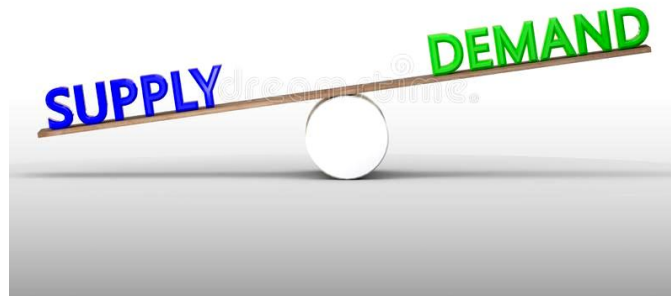
GCE ADVANCED LEVEL ENGLISH MEDIUM

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equilibrium determined by the demand and supply forces, perform an important duty within the society in producing and distributing goods and services.

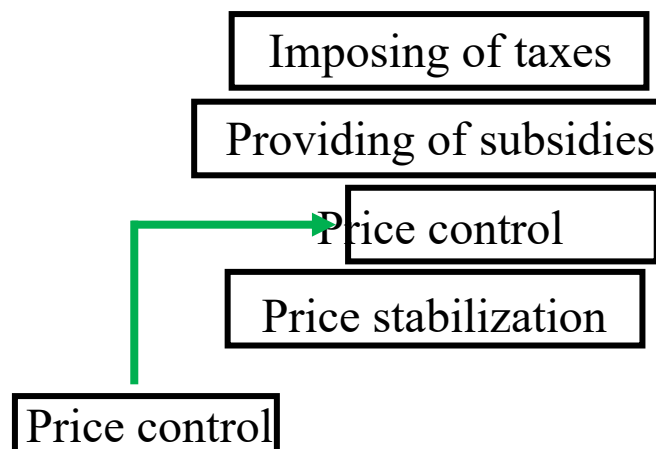


However when looking at market equilibrium from a social and political point of view some believe market equilibrium is as unfavorable to the society.



Because of this reason the government intervenes to the market in various ways.

#### ■ The main ways of government intervention ■



The prices of good or services which determined by the forces of demand and supply would not be fair to consumers and producers.

## In this situations government control the market price.

The price control is creating of an artificial price at the market by the government using rule and regulations.

Government aim is to protect producers and consumers by the price control.

### ►► Two types of price control.

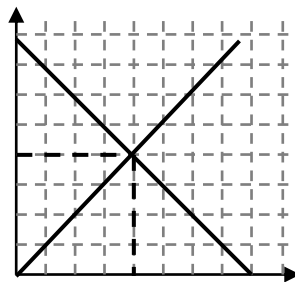
#### 1. Maximum price – ceiling price

#### 2. Minimum price – floor price

#### 1. Maximum price – ceiling price

When the government think that market equilibrium is unfair for consumers to give them fairness the legal price decided by the government is called maximum price.

Maximum price decided lower the equilibrium price is called as an effective maximum price.

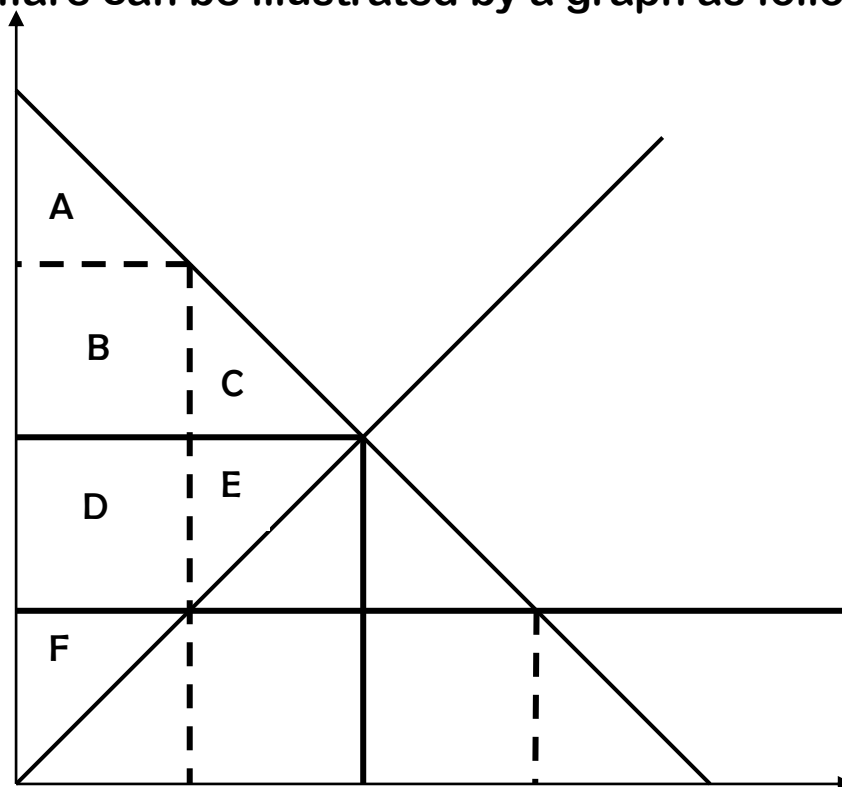


### ■ Effects of maximum price

- Creation of shortage of goods, due to the excess demand.
- Attempt to sell goods at an illegal black market price,
- Because of the black market price, selling of goods at a price more than the previous price.
- Due to the maximum price as consumer surplus and producer surplus are adversely affected it will also badly affects social welfare.

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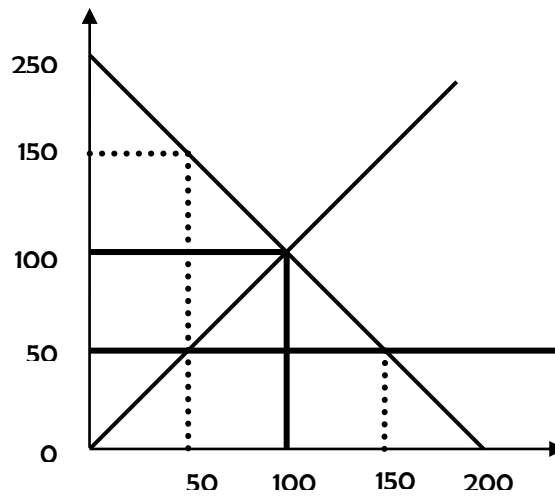
The effect of consumer and producer surplus and effect over welfare can be illustrated by a graph as follows



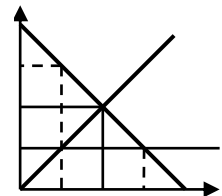
|                  | Before<br>Maximum price | After<br>Maximum price |
|------------------|-------------------------|------------------------|
| Consumer Surplus |                         |                        |
| Producer Surplus |                         |                        |
| Economic Surplus |                         |                        |

### ✪Exercise 1

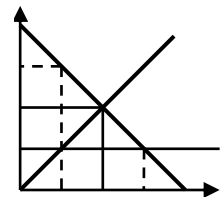
1 Consider a supply - demand equilibrium market below.



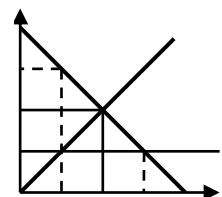
○ Calculate the consumer surplus before the maximum price



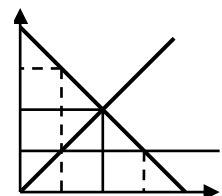
○ Calculate the producer surplus before the maximum price



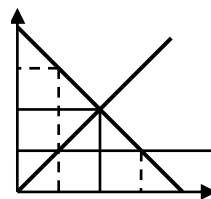
○ Calculate the economic surplus before the maximum price



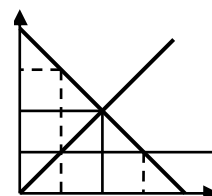
○ Find the Black market price after the maximum price



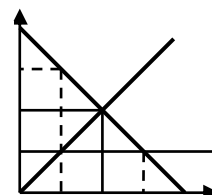
○ Find ED after the maximum price



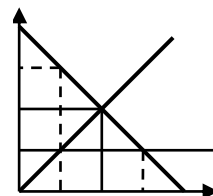
○ Find hidden amount by supplier after the maximum price



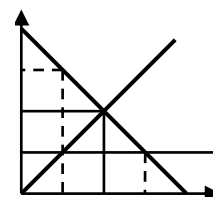
○ Find Black market revenue after the maximum price



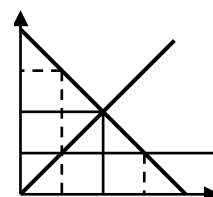
○ Find consumer surplus after the maximum price



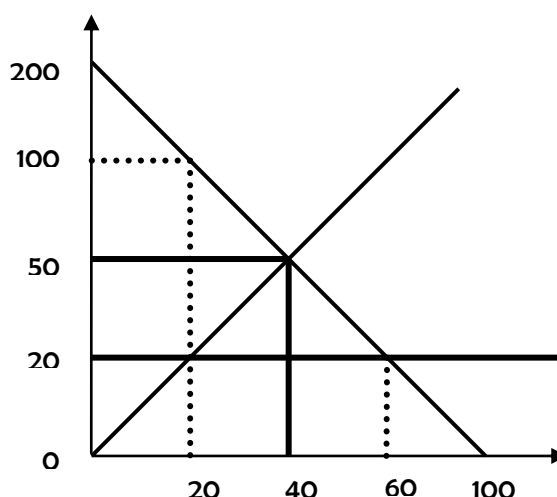
○ Find producer surplus after the maximum price



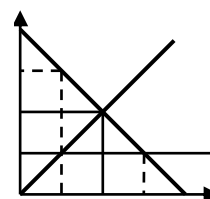
○ Find economic surplus after the maximum price



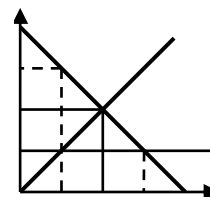
2 Consider a supply - demand equilibrium market below.



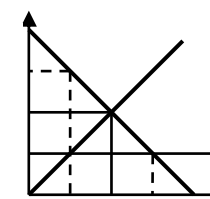
○ Calculate the consumer surplus before the maximum price



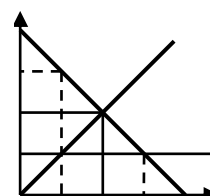
○ Calculate the producer surplus before the maximum price



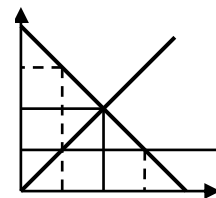
○ Calculate the economic surplus before the maximum price



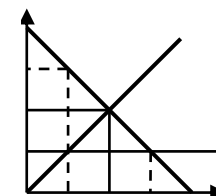
○ Find the Black market price after the maximum price



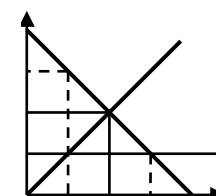
○ Find ED after the maximum price



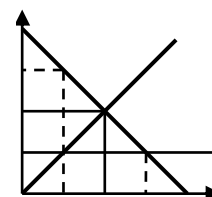
○ Find hidden amount by supplier after the maximum price



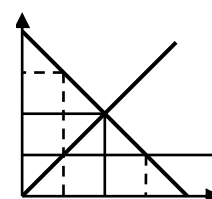
○ Find Black market revenue after the maximum price



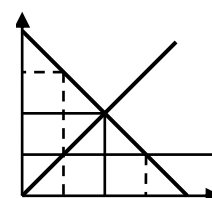
○ Find consumer surplus after the maximum price



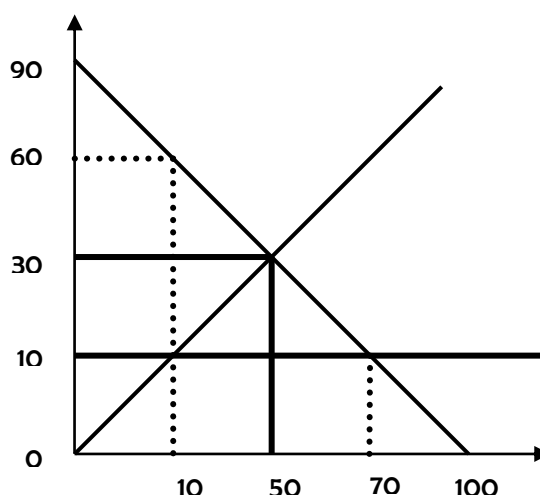
○ Find producer surplus after the maximum price



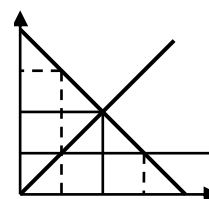
○ Find economic surplus after the maximum price



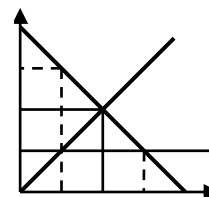
3 Consider a supply - demand equilibrium market below.



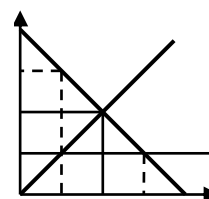
○ Calculate the consumer surplus before the maximum price



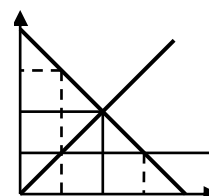
○ Calculate the producer surplus before the maximum price



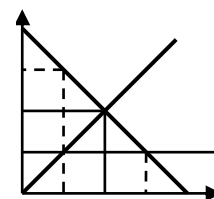
○ Calculate the economic surplus before the maximum price



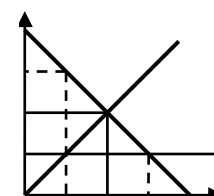
○ Find the Black market price after the maximum price



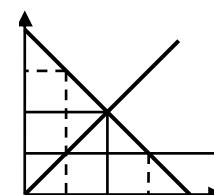
○ Find ED after the maximum price



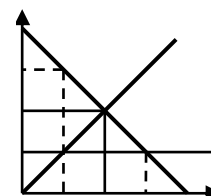
○ Find hidden amount by supplier after the maximum price



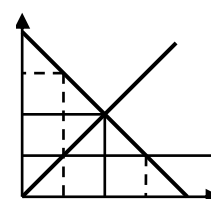
○ Find Black market revenue after the maximum price



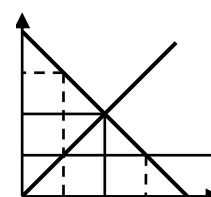
○ Find consumer surplus after the maximum price



○ Find producer surplus after the maximum price



○ Find economic surplus after the maximum price

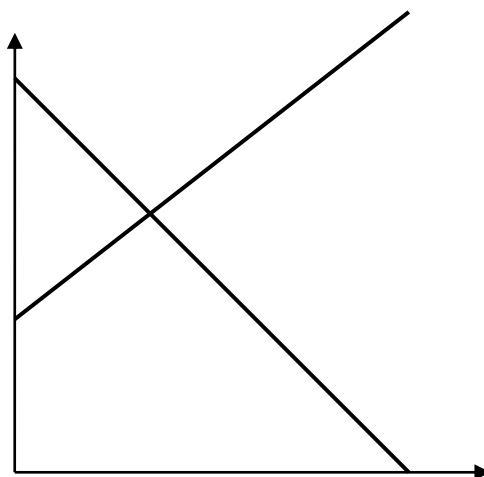


**✪Exercise 2****1)**

$$QD = 100 - 2P$$

$$QS = -60 + 3P$$

Draw a rough graph for the equations below



Calculate the equilibrium price and quantity

Find Maximum demand price and minimum supply price

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|--|--|

Calculate consumer surplus before the maximum price

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Calculate producer surplus before the maximum price

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Calculate economic surplus before the maximum price

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When the government sets a maximum price of 25 rupees, calculate the quantity of supply at that

When the government sets a maximum price of 25 rupees, calculate the quantity of demand at

Find Ed after the maximum price

Find hidden amount by supplier after the maximum price

Find black market price after the maximum price

Find black market revenue after the maximum price

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Calculate consumer surplus after the maximum price

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Calculate producer surplus after the maximum price

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Calculate economic surplus after the maximum price

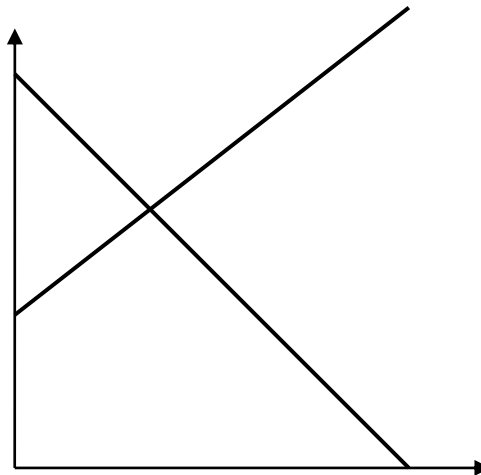
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2)

$$QD = 40 - 5P$$

$$QS = -10 + 5P$$

Draw a rough graph for the equations below



Calculate the equilibrium price and quantity

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Find Maximum demand price and minimum supply price

|  |  |
|--|--|
|  |  |
|--|--|

Calculate consumer surplus before the maximum price

Calculate producer surplus before the maximum price

Calculate economic surplus before the maximum price

When the government sets a maximum price of 3 rupees, calculate the quantity of supply at that price

When the government sets a maximum price of 3 rupees, calculate the quantity of demand at that price

Find Ed after the maximum price

Find hidden amount by supplier after the maximum price

Find black market price after the maximum price

Find black market revenue after the maximum price

Calculate consumer surplus after the maximum price

Calculate producer surplus after the maximum price

Calculate economic surplus after the maximum price

several approaches that can be taken to make maximum price meaningful.

### 1. Import

### 2. Providing incentives for the producers

### 3. Rationing

- Goods can be imported as a remedy to the shortage of the goods arise due to the excess demand created by the maximum price. By doing so the benefit of imposing a maximum price can be given to the consumer.
- Similarly by taking actions to encourage the producers who discouraged by the maximum price and by preventing market distortions the expected benefit has to be rationed among consumers.
- As the price mechanism being neutralized due to the maximum price policy scare, goods have to be rationed by using non price rationing techniques. Non price rationing techniques are as follows.

- Use of ration cards
- Queueing method
- Rationing with bribes
- Distribute connecting with other goods

## ■ Black market price

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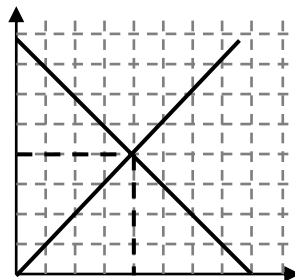
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### 1. Minimum Price – Floor price

When the government think that market equilibrium is unfair for producers to give them fairness the legal price decided by the government is called **minimum price**.

Minimum price decided higher than the equilibrium price is called as an **efficiency minimum price**.

Example : Implementing a minimum price for paddy implementing minimum wage rates.



### ■ Effects of minimum price

1. Accumulation of excess production of the market.
2. Unemployment problems can occur
3. Goods can be supplied to consumers at a discounted rate by keeping minimum price as nominal price. Sellers try escape from the law.
4. Excess investment situation can be occurred.

## The effect of consumer and producer surplus and effect

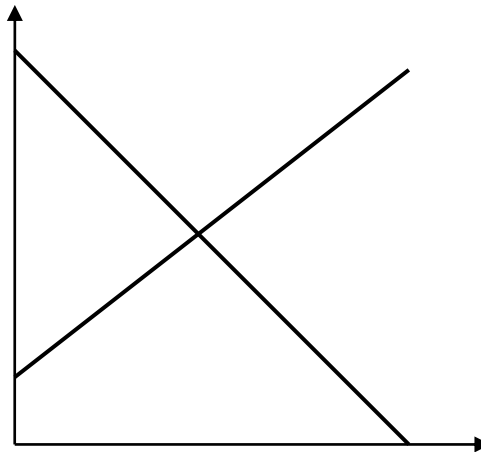
20 | Page

### ✪ Exercise 3

$$QD = 100 - 4P$$

$$QS = -50 + 6P$$

Draw a rough graph for the equations below



Calculate the equilibrium price and quantity

|  |  |
|--|--|
|  |  |
|--|--|

Find Maximum demand price and minimum supply price

|  |  |
|--|--|
|  |  |
|--|--|

Calculate consumer surplus before the minimum price

Calculate producer surplus before the minimum price

Calculate economic surplus before the minimum price

When the government sets a minimum price of 20 rupees, calculate the quantity of supply at that

When the government sets a minimum price of 20 rupees, calculate the quantity of demand at that price

Find Es after the maximum price

Calculate consumer surplus after the minimum price

Calculate producer surplus after the minimum price

Calculate economic surplus after the minimum price

\_\_\_\_\_To implement minimum price meaningfully some actions have to be taken. Following are some of such actions

**1. Storing of excess supply**

**2. By products**

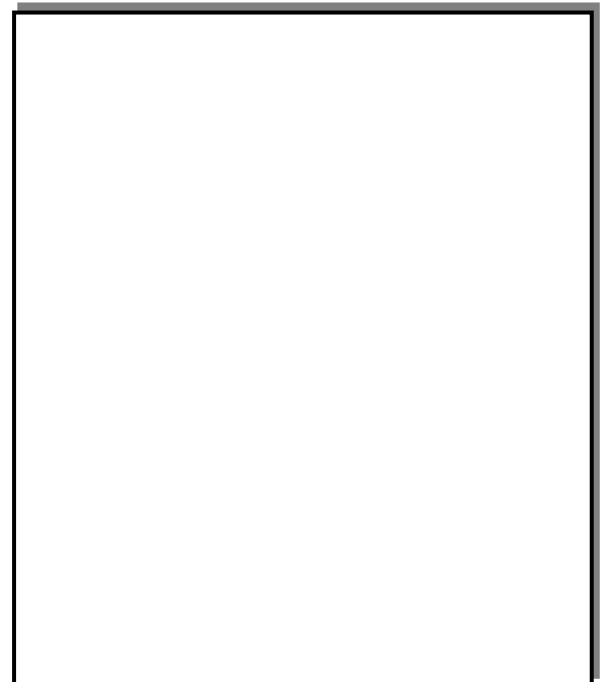
## 4. Exports

## 1. Price supporting government purchases

The graph shows a square divided into nine regions by a horizontal line, a vertical line, and two diagonal lines. The regions are labeled as follows:

- A**: Top-left triangle.
- B**: Top-left rectangle.
- C**: Top-right triangle.
- D**: Middle-left rectangle.
- E**: Middle-right triangle.
- F**: Bottom-left triangle.
- G**: Bottom-right triangle.
- H**: Bottom-right rectangle.
- I**: Top-right triangle.

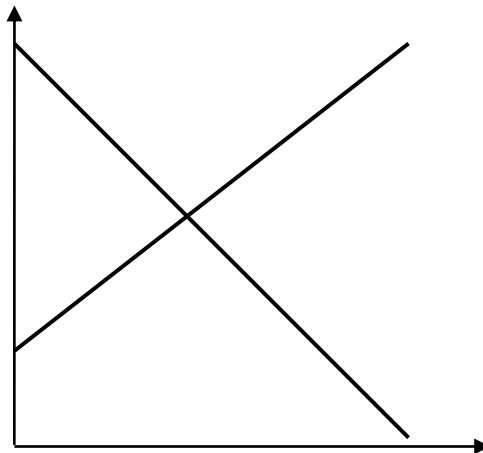
The regions are labeled with letters and the letter 'i'.



$$QD = 100 - 4P$$

$$QS = -50 + 6P$$

Draw a rough graph for the equations below



Calculate the equilibrium price and quantity

|  |
|--|
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Find Maximum demand price and minimum supply price

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When the government sets a maximum price of 20 rupees, calculate the ES at that price

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Calculate the cost of the government of purchasing the Excess supply

Calculate the consumer surplus after the government purchases the Excess supply

Calculate the producer surplus after the government purchases the Excess supply

Calculate the economic surplus after the government purchases the Excess supply

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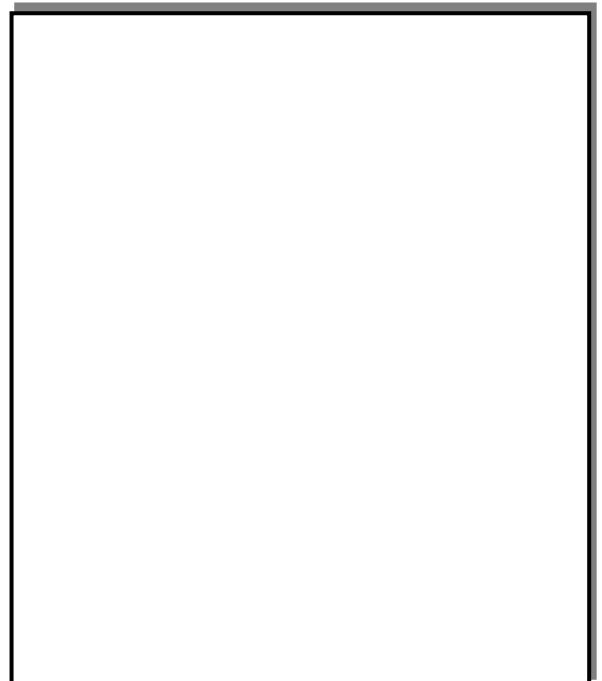
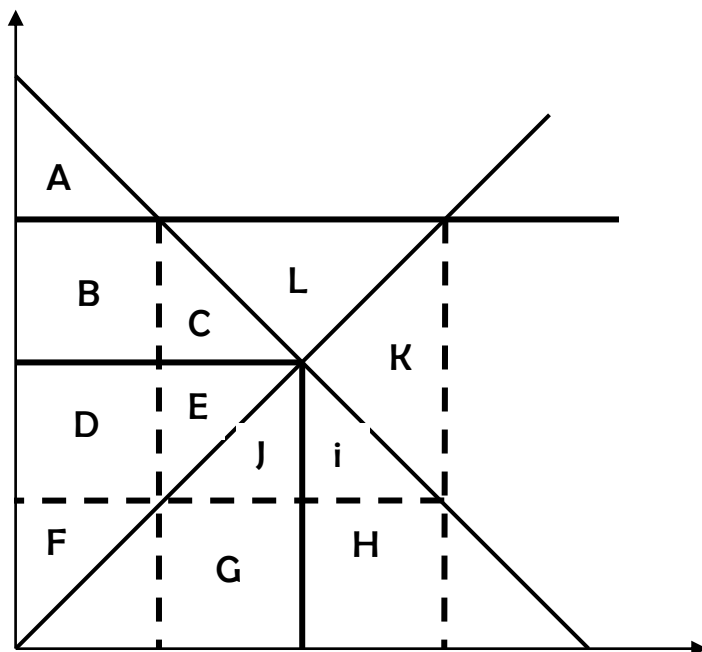


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## 2. Deficiency payment system

Deficiency payment system is the system in which government pay the difference between the minimum price imposed by the government and the price that the consumers are willing to pay to purchase the total supply that the producers are willing to supply at that minimum price.

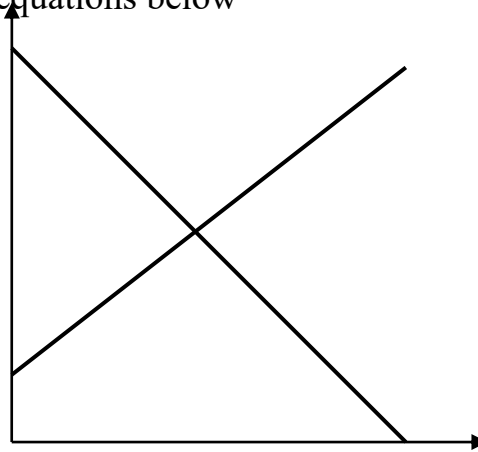
- Deficiency payment system can be illustrated by a graph as below



$$QD = 80 - 2P$$

$$QS = -20 + 2P$$

Draw a rough graph for the equations below



Calculate the equilibrium price and quantity

Calculate the equilibrium price and quantity consumer and producer surplus

When the government sets a maximum price of 30 rupees, calculate the ES at that price

When the government sets a maximum price of 30 rupees, find amount of the price that consumers willing to purchase the Quantity of supply at the minimum price

Calculate the deficiency payment for one unit

Calculate the total deficiency payment

Calculate the revenue of the producer

Calculate the consumer surplus after the deficiency payment

Calculate the producer surplus after the deficiency payment

Analyses government intervention in the free market through a unit tax.

**Imposing taxes on consumers and producers can be considered as a common situation of government intervention to the market process.**

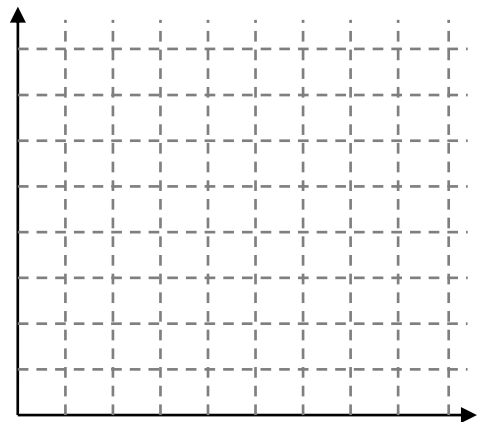
## ■ UNIT TAX

**Imposing taxes on a unit of a product produced or consumed is known as a unit tax or as a specific tax.**

Example.

- Imposing of Rs. 2 tax per one Kilogram of sugar.

| P | P+t | Qs |
|---|-----|----|
| 2 |     | 20 |
| 4 |     | 30 |
| 6 |     | 40 |
| 8 |     | 50 |



$$QS = a + b(p - t)$$

$Q_s = 100 + 20p$       **UNIT TAX 2**

$Q_s = -500 + 10p$       **UNIT TAX 2**

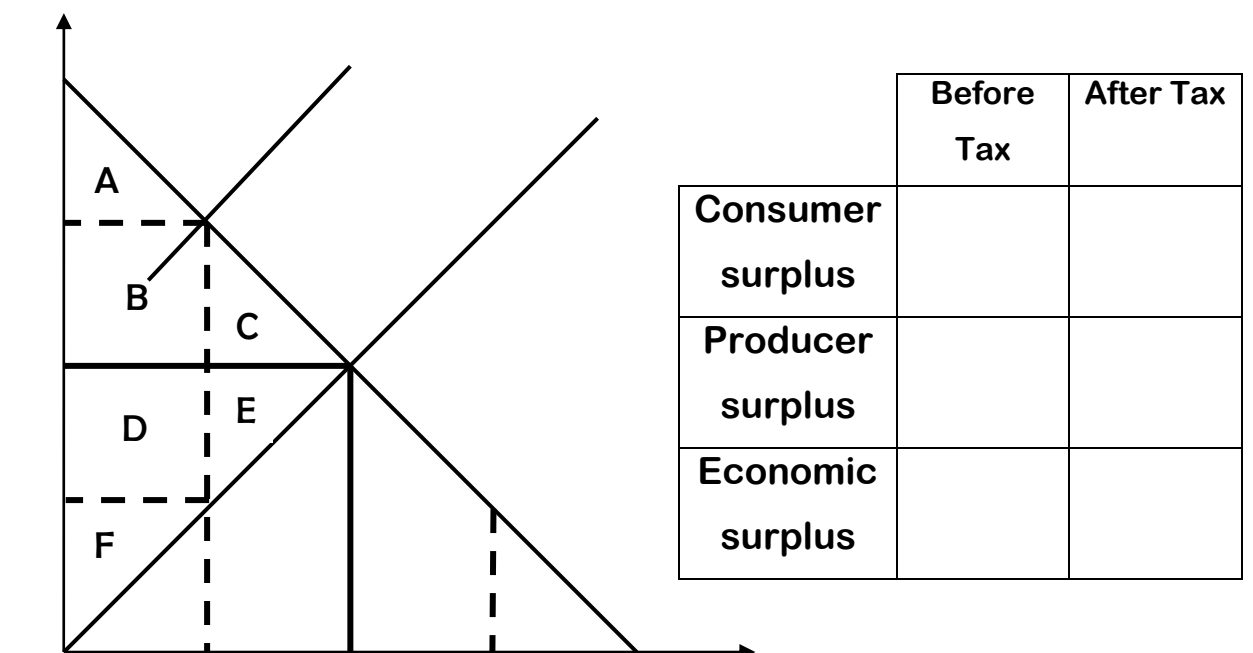
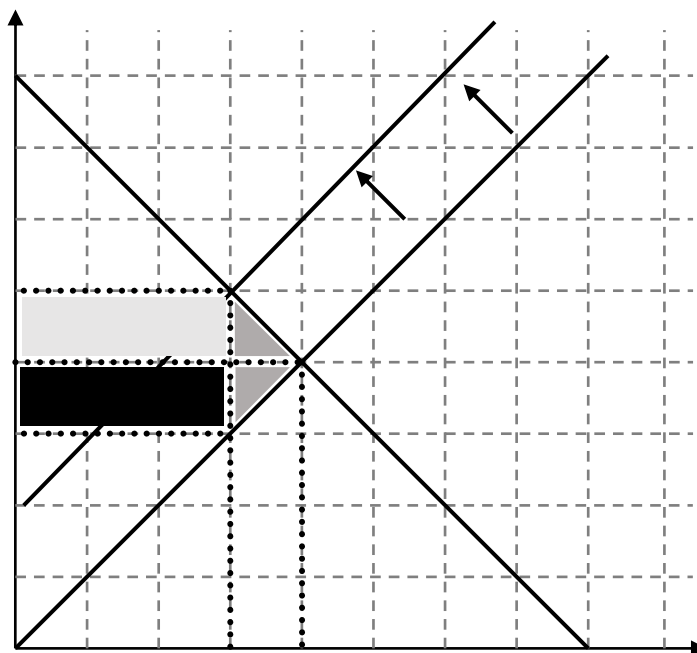
$Q_s = 300 + 2p$       **UNIT TAX 4**

$Q_s = -20 + 5p$       **UNIT TAX 2**

$Q_s = 250 + 10p$       **UNIT TAX 5**

$Q_s = -200 + 10p$       **UNIT TAX 4**

The effect of consumer and producer surplus and effect over welfare can be illustrated by a graph as follows

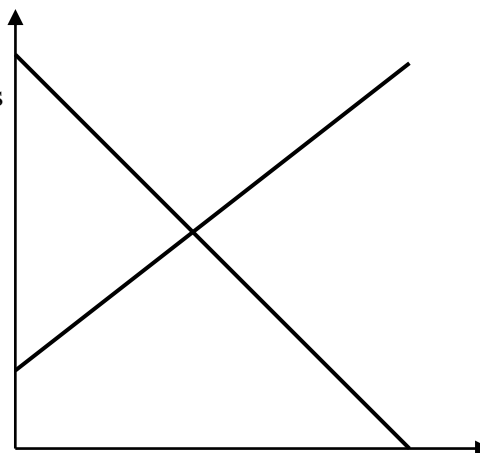


## EXERCISE 01

$$QD = 60 - 2p$$

$$QS = -10 + 3p$$

- 1) Draw a rough graph by using  $Q_s$   $Q_d$  equations



- 2) Calculate equilibrium price and quantity

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- 3) Calculate producer surplus and consumer surplus

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- 4) Calculate the new supply equation after the government imposes a tax of 5 rupees

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- 5) Calculate the equilibrium price and quantity after the unit tax

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- 6) Calculate the tax burden for producer

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7) Calculate the tax burden for consumer

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8) Calculate the producer tax burden

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9) Calculate the consumer tax burden

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10) Calculate the revenue received by the government due to the tax

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12) Calculate the consumer surplus after the tax

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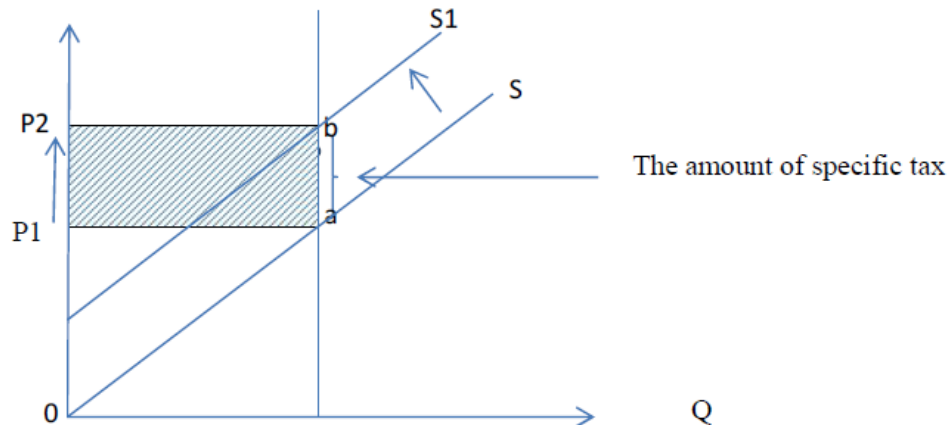
13) Calculate the producer surplus after the tax

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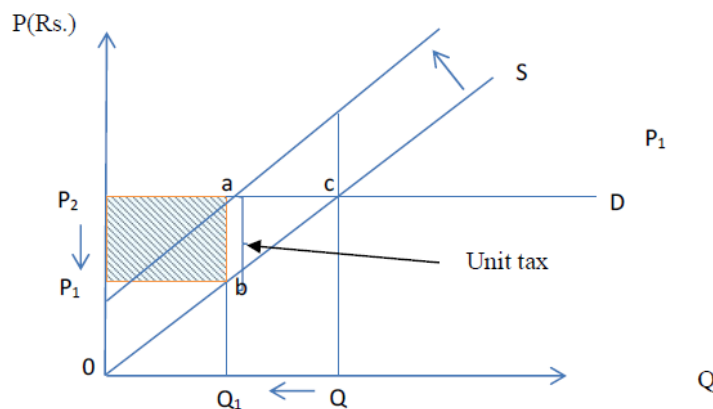
## Distribution of incidence of tax between consumer and producer based on the type of price elasticity of demand and supply

- Situation of perfectly inelastic demand:-



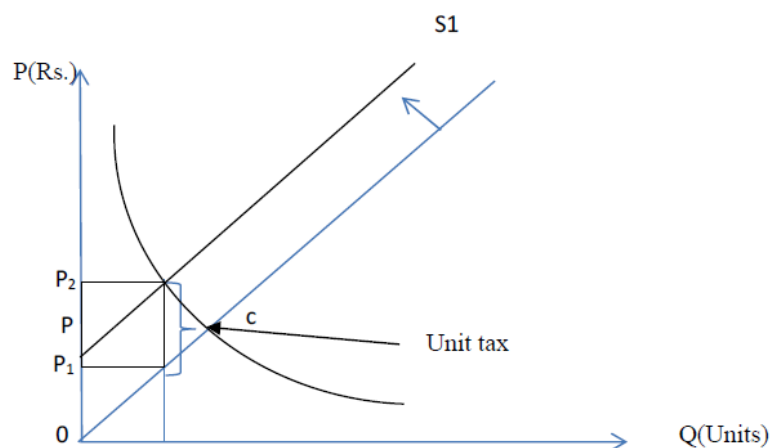
At a situation of perfectly inelastic demand increase in equilibrium price by the same amount of unit tax shows that the total burden of tax should be borne by the consumer.

Situation of perfectly elastic demand:-



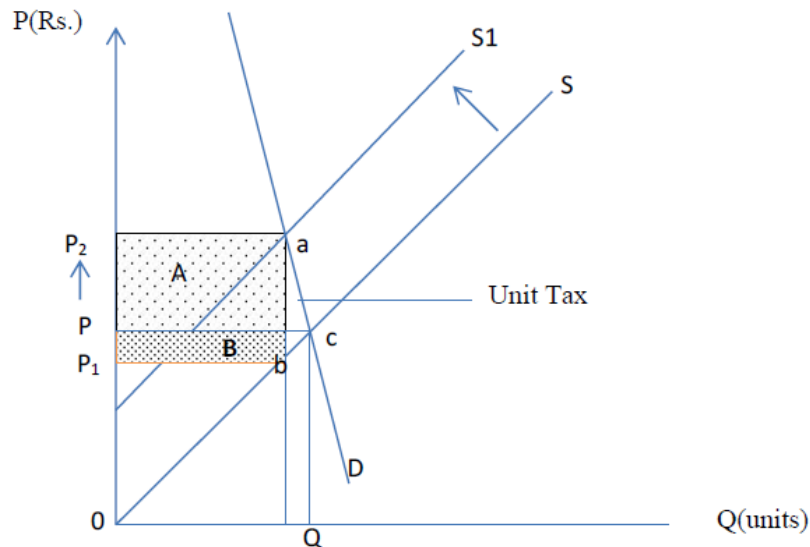
Therefore the area of tax burden shown by  $P_2 A B P_1$  will have to be borne by the producer.

Situation of unitary elastic demand:-



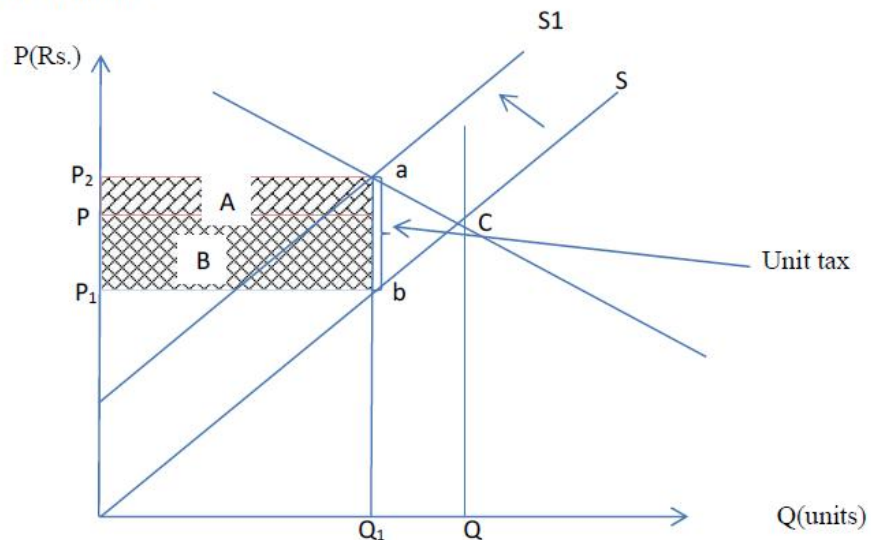
**In such a case the tax burden is shared equally between the producer and consumer**

- Situation of inelastic demand:-



**A situation of inelastic demand consumers bear most of the tax burden**

- Situation of elastic demand:-



**A situation of elastic demand producers bear most of the tax burden**

Analyses the government intervention to the market with providing of a unit subsidy ever supplier.

**Subsidy** is a transfer of money from the government to an entity. It leads to a fall in the price of the subsidised product. The objective of **subsidy** is to bolster the welfare of the society.

### ■ UNIT subsidy

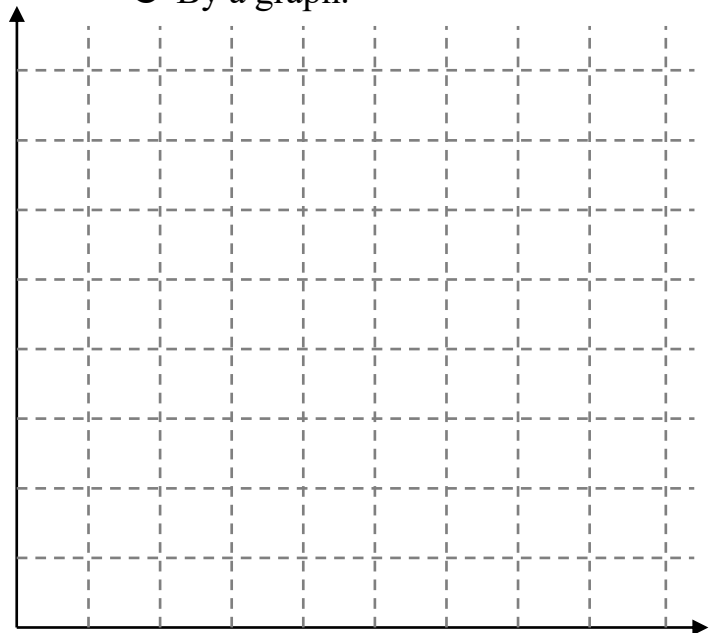
≈ A subsidy can also be called a negative tax.

○ The impacts of unit subsidies by the government can be expressed in several ways.

➡ By a supply schedule

| Price | Price after the subsidy රු 10.00 | QS  |
|-------|----------------------------------|-----|
| 0     | $0-10=$                          | -40 |
| 10    | $10-10=$                         | -20 |
| 20    | $20-10=$                         | 0   |
| 30    | $30-10=$                         | 20  |
| 40    | $40-10=$                         | 40  |
| 50    |                                  | 60  |
| 60    |                                  | 80  |

➡ By a graph.



Derive the supply equation

$$Q_d = a + b(p + s)$$

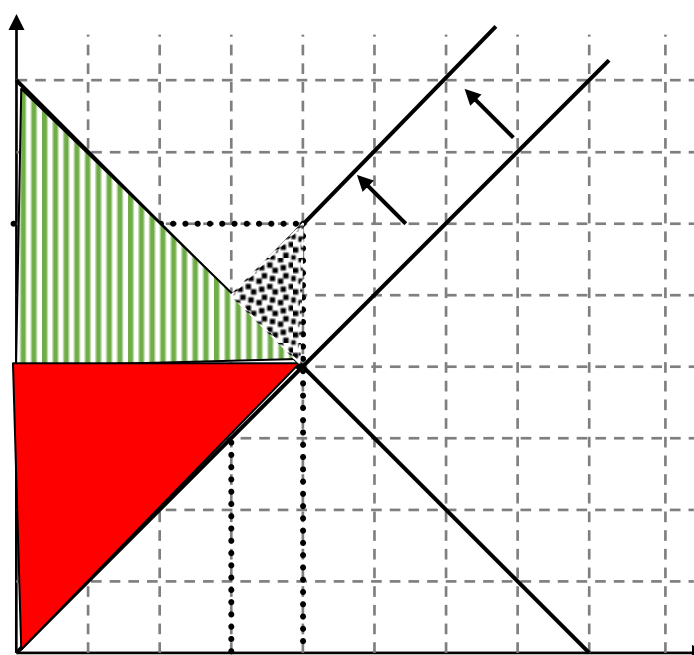
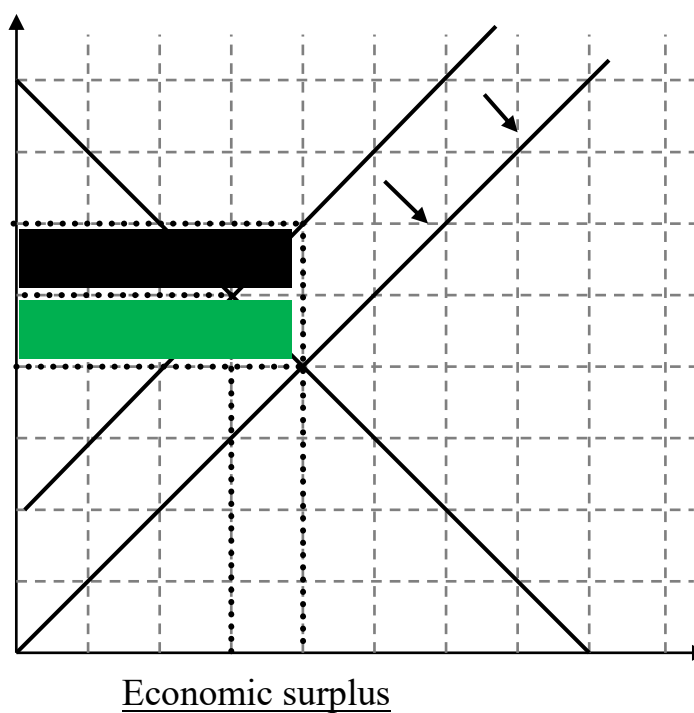
Q Draw the supply curves before and after the subsidy by considering the following information.

| P | P-s<br>subsidy = 2 | Qs | P  | P-s<br>subsidy = 10 | Qs  | P  | P-s<br>subsidy = 5 | Qs  |
|---|--------------------|----|----|---------------------|-----|----|--------------------|-----|
| 1 |                    | 20 | 20 |                     | 100 | 5  |                    | 25  |
| 2 |                    | 30 | 40 |                     | 200 | 10 |                    | 50  |
| 3 |                    | 40 | 60 |                     | 300 | 15 |                    | 75  |
| 4 |                    | 50 | 80 |                     | 400 | 20 |                    | 100 |

Q Derive the supply equation after the subsidy by considering the following information

|                                                              |                                                            |
|--------------------------------------------------------------|------------------------------------------------------------|
| <div> <math>Q_s = 100 + 20p</math>      subsidy = 10 </div>  | <div> <math>Q_s = 100 + 30p</math>      subsidy = 2 </div> |
| <div> <math>Q_s = -500 + 10p</math>      subsidy = 20 </div> | <div> <math>Q_s = -100 + 5p</math>      subsidy = 5 </div> |

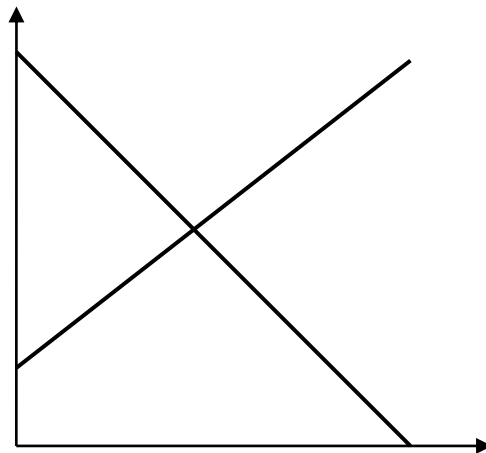
### Cost of government due to subsidy



**OX 01**

$$QD = 60 - 2p$$

$$QS = -10 + 3p$$



1) Draw a rough graph by using above details.

2) Calculate the equilibrium price and quantity.

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3) Calculate the producer and consumer surplus at the equilibrium.

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4) The government provides a unit subsidy of Rs. 5.00 for the producers in this market. Under this situation calculate the supply equation after the subsidy.

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5) Calculate the neq equilibrium price and quantity after the unit subsidy.

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6) How many units received by the consumer from the subsidy?

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7) How many units received by the consumer from the subsidy?

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8) Calculate the total subsidy benefit of the producer.

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9) Calculate the total subsidy benefit of the consumer

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12) Calculate the consumer surplus after the unit subsidy

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13) Calculate the producer surplus after the unit subsidy

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